



Koneru Lakshmaiah Education Foundation

(Deemed to be University estd. u/s. 3 of the UGC Act, 1956)

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DEPARTMENT OF CSE

PROJECT ABSTRACT SUBMISSION

COURSE: ENGINEERING CAPSTONE PROJECT – 1(23IE4053R/23IE4053A) A.Y. 2026 - 2027

TITLE OF THE PROJECT		
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Abstract

The **MarketPulse – Stock Return Distribution & Risk Analytics System** is a Data Science-based analytical platform designed to study historical stock market behavior and evaluate investment risk using statistical techniques and data visualization. The system collects and processes historical stock price data to calculate daily returns, volatility, mean returns, standard deviation, skewness, and kurtosis. It analyzes the distribution of stock returns to identify trends, variations, and potential risk patterns.

The proposed system provides an interactive dashboard where users can select stocks, analyze historical performance, visualize return distributions, compare multiple stocks, and identify periods of high volatility. Statistical and graphical techniques such as histograms, box plots, correlation analysis, and moving averages are used to present the results in an understandable manner. The system can also estimate the probability of positive or negative returns based on historical observations.

The project demonstrates the practical application of **Python, Pandas, NumPy, statistical analysis, data visualization, and Streamlit** in financial data science. The main objective is to transform complex historical market data into meaningful insights that can help users better understand stock performance and associated risk. The system is intended as an **analytical and educational tool**, rather than a system for guaranteed stock-price prediction or investment advice.

Signature of the Guide

HOD

